

# Chemical Bonding and Molecular Geometry

## Bonding, Lewis Structures, and VSEPR Theory

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# Outline

1. Ionic Bonding
2. Covalent Bonding
3. Lewis Symbols and Structures
4. Formal Charges and Resonance
5. Strengths of Ionic and Covalent Bonds
6. Molecular Structure and Polarity

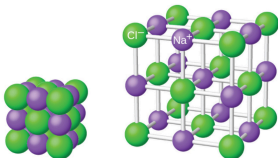
# Ionic Bonding

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# Ionic bonds involve electrostatic attraction

## Definition

**Ionic bonds** involve the electrostatic attraction of oppositely charged ions in accordance with Coulomb's law.



Coulomb's law  $E = k_e e^2 \frac{q_+ q_-}{r}$

Lattice energy  $\text{NaCl(s)} \longrightarrow \text{Na}^{\text{+}}(\text{g}) + \text{Cl}^{\text{-}}(\text{g}), \Delta H^{\circ} = 769 \text{ kJ}$

# Ionic charges follow patterns

Most main-group cations are derived from loss of all valence electrons to form the configuration of the nearest noble gas.



Most main-group anions follow a similar pattern.



# Covalent Bonding

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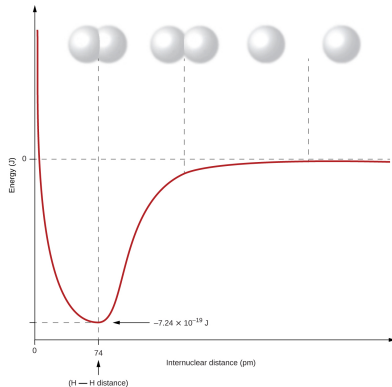
# Covalent bonding involves electron sharing

## Definition

**Covalent bonds** result from mutual attraction of atoms for a shared pair of electrons.

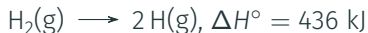
## Definition

The interatomic distance at which energy is minimized is the **bond length**.

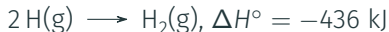


## Bound atoms are more stable than free atoms

Cleavage of covalent bonds is always an endothermic process.



Formation of covalent bonds releases energy.

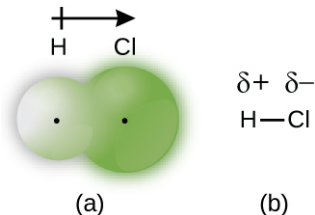




# Electrons are often shared unequally

## Definition

Atoms of different electronegativity share electrons unequally in a **polar covalent bond**.



**Figure 1:** The covalent bond in HCl is polar.

# Electronegativity points to bond polarity

## Definition

**Electronegativity** is the tendency of an atom to attract electrons to itself. It determines how electrons are shared in covalent bonds.

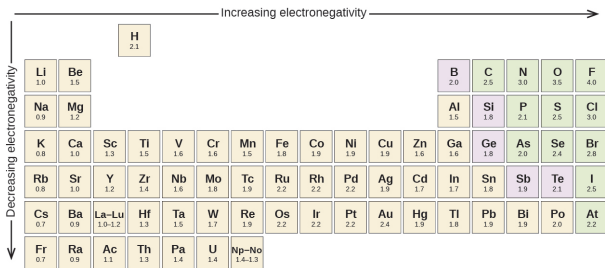
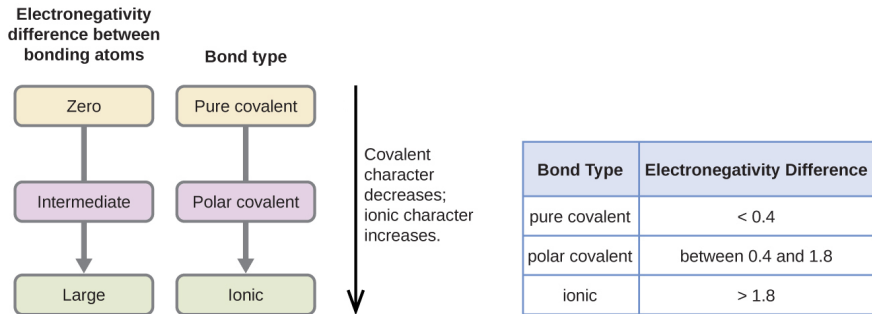


Figure 2: Electronegativity exhibits an important periodic trend.

# Electronegativity points to bond polarity



**Figure 3:** The ionic or covalent nature of a bond depends on the electronegativity difference between the atoms involved.

# Electronegativity points to bond polarity

## Electronegativity and Bond Polarity

Arrange the following covalent bonds in order of increasing polarity. Then designate the positive and negative atoms in the bonds using the symbols  $\delta+$  and  $\delta-$ : C-H, C-N, C-O, N-H, O-H, S-H.

# Lewis Symbols and Structures

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# Lewis symbols depict atomic valence electrons and nuclei

## Definition

A **Lewis symbol** consists of an elemental symbol surrounded by one dot for each of its valence electrons.

| Atoms      | Electronic Configuration | Lewis Symbol                                                                                                                                |
|------------|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| sodium     | $[\text{Ne}]3s^1$        | $\text{Na} \cdot$                                                                                                                           |
| magnesium  | $[\text{Ne}]3s^2$        | $\cdot \text{Mg} \cdot$                                                                                                                     |
| aluminum   | $[\text{Ne}]3s^2 3p^1$   | $\cdot \overset{\cdot}{\underset{\cdot}{\text{Al}}} \cdot$                                                                                  |
| silicon    | $[\text{Ne}]3s^2 3p^2$   | $\cdot \overset{\cdot}{\underset{\cdot}{\underset{\cdot}{\text{Si}}}} \cdot$                                                                |
| phosphorus | $[\text{Ne}]3s^2 3p^3$   | $\cdot \overset{\cdot}{\underset{\cdot}{\underset{\cdot}{\underset{\cdot}{\text{P}}}}} \cdot$                                               |
| sulfur     | $[\text{Ne}]3s^2 3p^4$   | $\overset{\cdot}{\underset{\cdot}{\underset{\cdot}{\underset{\cdot}{\underset{\cdot}{\text{S}}}}} \cdot$                                    |
| chlorine   | $[\text{Ne}]3s^2 3p^5$   | $\overset{\cdot}{\underset{\cdot}{\underset{\cdot}{\underset{\cdot}{\underset{\cdot}{\underset{\cdot}{\text{Cl}}}}} \cdot$                  |
| argon      | $[\text{Ne}]3s^2 3p^6$   | $\overset{\cdot}{\underset{\cdot}{\underset{\cdot}{\underset{\cdot}{\underset{\cdot}{\underset{\cdot}{\underset{\cdot}{\text{Ar}}}}} \cdot$ |

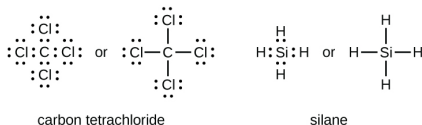
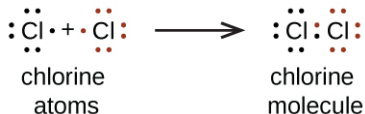
# Lewis structures show electrons and nuclei in molecules

## Definition

**Lewis structures** describe the bonding in molecules and polyatomic ions.

## Definition

The tendency of main-group atoms to form enough bonds to obtain eight valence electrons is known as the **octet rule**.



# Multiple electron pairs can be shared

When atoms share two (three) electron pairs, a **double (triple) bond** is formed.

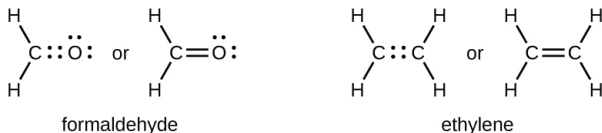


Figure 4: Examples of double bonds in Lewis structures.



Figure 5: Examples of triple bonds in Lewis structures.



# Drawing a Lewis structure using a given molecular formula

1. Determine the total number of valence electrons. For cations, subtract one electron for each positive charge. For anions, add one electron for each negative charge.
2. Draw a skeletal structure of the molecule or ion, arranging the atoms around a central atom. (Generally, the least electronegative element should be placed in the center.) Connect each atom to the central atom with a single bond (one electron pair).
3. Distribute the remaining electrons as lone pairs on the terminal atoms (except hydrogen), completing an octet around each atom.
4. Place all remaining electrons on the central atom.
5. Rearrange the electrons of the outer atoms to make multiple bonds with the central atom in order to obtain octets wherever possible.

# Practice drawing Lewis structures

## Drawing Lewis Structures

Draw a Lewis structure for  $\text{SiH}_4$ .

# Practice drawing Lewis structures

## Drawing Lewis Structures

Draw a Lewis structure for  $\text{CHO}_2^-$ .

# Practice drawing Lewis structures

## Drawing Lewis Structures

Draw a Lewis structure for  $\text{NO}^+$ .

# The octet rule has many exceptions

Odd-electron molecules NO, CH<sub>3</sub>

Electron-deficient molecules BF<sub>3</sub>, CH<sub>3</sub><sup>+</sup>

Hypervalent molecules IF<sub>5</sub>, SF<sub>6</sub>

# Formal Charges and Resonance

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# Formal charge shows a lack or excess of electrons

## Definition

**Formal charge** is the hypothetical charge an atom would have if bonding electrons were split evenly between linked atoms.

$$FC = VE(\text{free atom}) - NBE - \frac{1}{2}BE$$

## Conservation of Charge

The sum of formal charges in a molecule must equal the overall charge of the molecule.

# Include formal charges in Lewis structures

## Calculating Formal Charges

Assign formal charges to each atom in the interhalogen ion  $\text{ICl}_4^-$ .



# Formal charges guide Lewis structures

1. A molecular structure in which all formal charges are zero is preferable to one in which some formal charges are not zero.
2. If the structure must have nonzero formal charges, the arrangement with the smallest nonzero formal charges is preferable.
3. Lewis structures are preferable when adjacent formal charges are zero or of opposite sign.
4. When we must choose among several structures with similar distributions of formal charges, the structure with negative formal charges on more electronegative atoms is preferable.

# Formal charges guide Lewis structures

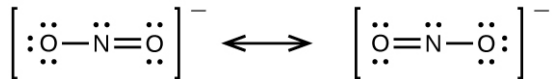
## Using Formal Charge to Determine Structure

Nitrous oxide,  $\text{N}_2\text{O}$ , commonly known as laughing gas, is used as an anesthetic in minor surgeries, such as the routine extraction of wisdom teeth. What is the structure of nitrous oxide?

# Resonance structures capture electron delocalization

## Definition

Molecules containing delocalized electrons require multiple **resonance structures** for complete description. The true electron distribution is an average of those in the resonance forms.

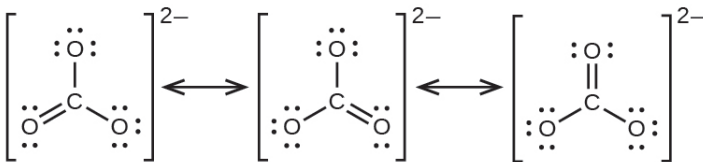


**Figure 6:** Nitrite anion is characterized by two significant resonance forms.

# Resonance structures capture electron delocalization

## Definition

The true structure derived from averaging of resonance forms is called the **resonance hybrid**.



**Figure 7:** Carbonate anion has three significant resonance forms.

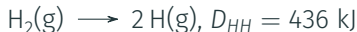
## Strengths of Ionic and Covalent Bonds

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# Energy is required to break bonds

## Definition

**Bond dissociation energy (BDE)** is the energy required to separate the atoms in a bond.



- Bond strength follows bond order: triple > double > single
- More polar bonds are generally stronger than less polar bonds:  
 $C-F > C-O > C-C$

## $\Delta H_{rxn}^\circ$ can be approximated using BDEs

### Calculating Enthalpy Change Using Bonds Made and Broken

Any reaction can be envisioned as a series of bond cleavages and bond formations. The enthalpy change is approximated by the difference between BDEs for the bonds broken (positive) and bonds made (negative).

$$\Delta H^\circ = \sum_i^{broken} D_i - \sum_j^{made} D_j$$

$\Delta H_{rxn}^{\circ}$  can be approximated using BDEs

### Bond Energies and Reaction Enthalpy Change

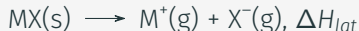
Using bond energies, calculate the approximate enthalpy change for the reaction:  $\text{CO(g)} + 2 \text{H}_2\text{(g)} \longrightarrow \text{CH}_3\text{OH(g)}$ .



# Ionic bonds are much stronger than covalent bonds

## Definition

**Lattice energy** is the energy required to separate 1 mol of a solid ionic lattice into gaseous ions.



Trends in lattice energy can be rationalized using Coulomb's law.

$$\Delta H_{\text{lat}} = \left| C \frac{Z^+ Z^-}{R_0} \right|$$

E.g.,  $\text{MgO} > \text{MgF}_2 > \text{MgI}_2 > \text{LiF}$

## Relative lattice energies are based on radius and charge

### Comparing Lattice Energies

Which has the larger lattice energy,  $\text{Al}_2\text{O}_3$  or  $\text{Al}_2\text{Se}_3$ ?

# Born-Haber cycles are an application of Hess's law

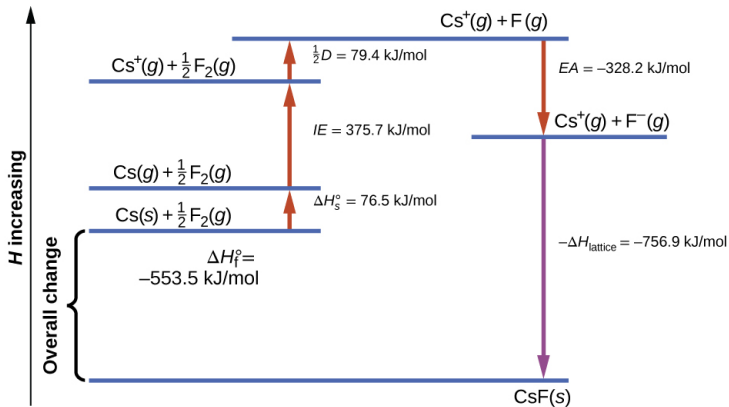


Figure 8: Born-Haber cycle for cesium fluoride.

## Born-Haber cycles are an application of Hess's law

The cycle starts and ends at CsF(s). If any six of the enthalpy changes or their reverses are known, the seventh can be calculated using Hess's law.

$$\Delta H_{lat} = 76.5 \text{ kJ mol}^{-1} + 79.4 \text{ kJ mol}^{-1} + 375.7 \text{ kJ mol}^{-1} + (-328.2 \text{ kJ mol}^{-1}) - (-553.5 \text{ kJ mol}^{-1}) = 756.9 \text{ kJ mol}^{-1}$$

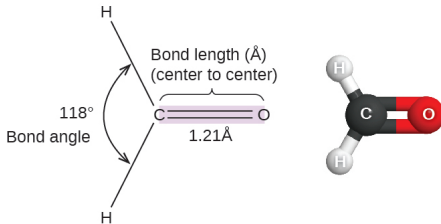
# Molecular Structure and Polarity

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# Geometries are described by bond lengths and angles

## Definition

A **bond angle** is the angle between any two bonds sharing a common atom. A **bond length** is the distance between the nuclei of two bonded atoms along the axis joining the nuclei.



**Figure 9:** Bond lengths and angles in formaldehyde,  $\text{CH}_2\text{O}$ .

# VSEPR theory enables prediction of geometry

## Definition

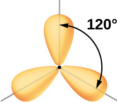
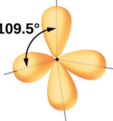
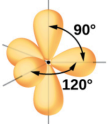
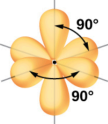
**Valence shell electron-pair repulsion (VSEPR) theory** posits that repulsion of electron pairs gives rise to molecular geometry.

**Steric number** Count of contiguous regions of electron density around the central atom

**Electron-pair geometry** Arrangement of regions of electron density around the central atom based on VSEPR

**Molecular geometry** Arrangement of atoms around the central atom, disregarding nonbonding pairs

# Survey of electron-pair geometries

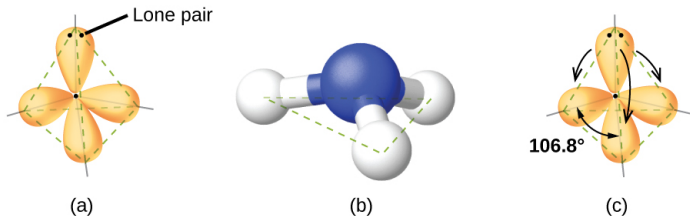
| Number of regions        | Two regions of high electron density (bonds and/or unshared pairs)                | Three regions of high electron density (bonds and/or unshared pairs)              | Four regions of high electron density (bonds and/or unshared pairs)               | Five regions of high electron density (bonds and/or unshared pairs)                | Six regions of high electron density (bonds and/or unshared pairs)                  |
|--------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Spatial arrangement      |  |  |  |  |  |
| Line-dash-wedge notation | <chem>H-Be-H</chem>                                                               |  |  |  |  |



# Molecular geometry is a distinct concept

## An Important Distinction

Nonbonding lone pairs cause the electron-pair geometry and molecular geometry to differ. They are not the same in general!

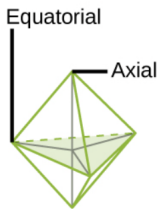


**Figure 10:** (a) Electron-pair geometry of ammonia. (b) Molecular geometry of ammonia. (c) Deviations from idealized VSEPR bond angles.

# Survey of molecular geometries

| Number of electron regions | Electron region geometries:<br>0 lone pair                                                                | 1 lone pair                                                                                             | 2 lone pairs                                                                                         | 3 lone pairs                                                                                 | 4 lone pairs                                                                                 |
|----------------------------|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| 2                          | <br>Linear               |                                                                                                         |                                                                                                      |                                                                                              |                                                                                              |
| 3                          | <br>Trigonal planar      | <br>Bent or angular    |                                                                                                      |                                                                                              |                                                                                              |
| 4                          | <br>Tetrahedral          | <br>Trigonal pyramidal | <br>Bent or angular |                                                                                              |                                                                                              |
| 5                          | <br>Trigonal bipyramidal | <br>Sawhorse or seesaw | <br>T-shape         | <br>Linear  |                                                                                              |
| 6                          | <br>Octahedral           | <br>Square pyramidal   | <br>Square planar   | <br>T-shape | <br>Linear |

# Distinct positions exist in the TBP geometry



## Example

Consider the structure of  $\text{ClF}_3$ .

**Figure 11:** The trigonal bipyramidal geometry has two distinct positions called **axial** and **equatorial**. These positions are energetically distinct.

## Apply VSEPR theory to predict molecular geometry

1. Write the Lewis structure of the molecule or polyatomic ion.
2. Determine the steric number of the central atom. A single, double, or triple bond counts as one region of electron density.
3. Identify the electron-pair geometry based on the number of regions of electron density: linear, trigonal planar, tetrahedral, trigonal bipyramidal, or octahedral.
4. Use the number of lone pairs to determine the molecular geometry. If more than one geometry is possible, choose the one that minimizes electron-electron repulsions.

# Apply VSEPR theory to predict molecular geometry

## Predicting Molecular Geometry

Predict the electron-pair geometry and molecular structure (molecular geometry) of the  $\text{SF}_4$  molecule.

# Apply VSEPR theory to predict molecular geometry

## Predicting Molecular Geometry

Predict the electron-pair geometry and molecular structure (molecular geometry) of the  $\text{XeF}_4$  molecule.

# Dipole moment is a vector that depicts polarity

## Definition

Bonds between atoms of different electronegativity have a **bond dipole moment**  $\mu = Qr$ . Dipoles are represented as vectors.

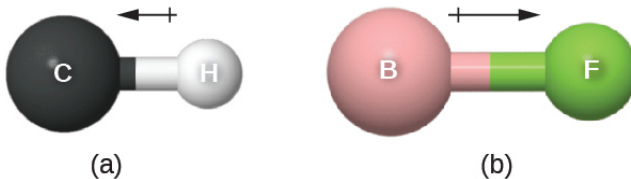


Figure 12: Bond dipoles of (a) C—H and (b) B—F bonds.

# Molecular dipole is a vector sum of bond dipoles

## Definition

**Molecular dipole moment**, the vector sum of the bond dipoles, reflects the extent of charge separation in the molecule as a whole.

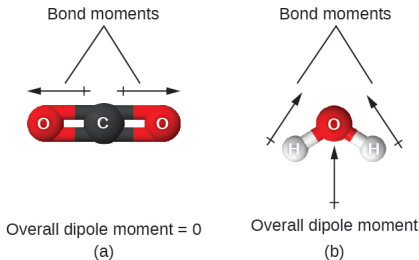
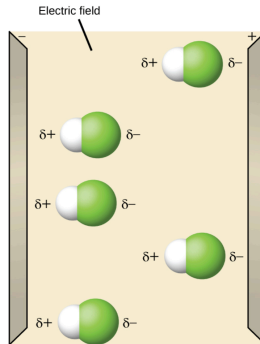


Figure 13: Molecular dipoles of (a) **nonpolar**  $\text{CO}_2$  and (b) **polar**  $\text{H}_2\text{O}$ .



# Molecular polarity affects macroscopic properties

- Molecules of similar polarity tend to form solutions (like dissolves like).
- Polar molecules are more strongly affected by electric fields than nonpolar molecules.



**Figure 14:** HCl molecules align with an applied electric field.

# Experiment with polarity

## Simulations of Polarity

Open the [molecule polarity](#) simulation and select the "Three Atoms" tab at the top. Use the electronegativity controls to determine how the molecular dipole will look if (a) A and C are very electronegative and B is in the middle of the range.

# Experiment with polarity

## Simulations of Polarity

Open the [molecule polarity](#) simulation and select the "Three Atoms" tab at the top. Use the electronegativity controls to determine how the molecular dipole will look if (b) A is very electronegative while B and C are not.